

1. Create numeric, character, and logical vectors and display type and content.

```
R Console
>
> # Display numeric vector and its type
> print("Numeric Vector:")
[1] "Numeric Vector:"
> print(numeric_vector)
[1] 10 20 30 40 50
> print(typeof(numeric_vector))
[1] "double"
>
> # Display character vector and its type
> print("Character Vector:")
[1] "Character Vector:"
> print(character_vector)
[1] "Apple" "Banana" "Cherry"
> print(typeof(character_vector))
[1] "character"
>
> # Display logical vector and its type
> print("Logical Vector:")
[1] "Logical Vector:"
> print(logical_vector)
[1] TRUE FALSE TRUE FALSE
> print(typeof(logical_vector))
[1] "logical"
>

Untitled - R Editor
# Create a numeric vector
numeric_vector <- c(10, 20, 30, 40, 50)

# Create a character vector
character_vector <- c("Apple", "Banana", "Cherry")

# Create a logical vector
logical_vector <- c(TRUE, FALSE, TRUE, FALSE)

# Display numeric vector and its type
print("Numeric Vector:")
print(numeric_vector)
print(typeof(numeric_vector))

# Display character vector and its type
print("Character Vector:")
print(character_vector)
print(typeof(character_vector))

# Display logical vector and its type
print("Logical Vector:")
print(logical_vector)
print(typeof(logical_vector))
```

2. Create labeled matrices (5x4, 3x3, 2x2) filled by row/column.

```
R Console
> # 3 x 3 Matrix (Filled by Column)
> matrix2 <- matrix(1:9, nrow = 3, ncol = 3, byrow = FALSE)
> rownames(matrix2) <- paste("R", 1:3, sep = "")
> colnames(matrix2) <- paste("C", 1:3, sep = "")
>
> print("3x3 Matrix (Filled by Column):")
[1] "3x3 Matrix (Filled by Column):"
> print(matrix2)
      C1 C2 C3
R1  1  4  7
R2  2  5  8
R3  3  6  9
>
> # 2 x 2 Matrix (Filled by Row)
> matrix3 <- matrix(1:4, nrow = 2, ncol = 2, byrow = TRUE)
> rownames(matrix3) <- paste("R", 1:2, sep = "")
> colnames(matrix3) <- paste("C", 1:2, sep = "")
>
> print("2x2 Matrix (Filled by Row):")
[1] "2x2 Matrix (Filled by Row):"
> print(matrix3)
      C1 C2
R1  1  2
R2  3  4
>

D:\study\R programming\programs\Create numeric, character, and logical vectors and display ty...
# 5 x 4 Matrix (Filled by Row)
matrix1 <- matrix(1:20, nrow = 5, ncol = 4, byrow = TRUE)
rownames(matrix1) <- paste("R", 1:5, sep = "")
colnames(matrix1) <- paste("C", 1:4, sep = "")

print("5x4 Matrix (Filled by Row):")
print(matrix1)

# 3 x 3 Matrix (Filled by Column)
matrix2 <- matrix(1:9, nrow = 3, ncol = 3, byrow = FALSE)
rownames(matrix2) <- paste("R", 1:3, sep = "")
colnames(matrix2) <- paste("C", 1:3, sep = "")

print("3x3 Matrix (Filled by Column):")
print(matrix2)

# 2 x 2 Matrix (Filled by Row)
matrix3 <- matrix(1:4, nrow = 2, ncol = 2, byrow = TRUE)
rownames(matrix3) <- paste("R", 1:2, sep = "")
colnames(matrix3) <- paste("C", 1:2, sep = "")

print("2x2 Matrix (Filled by Row):")
print(matrix3)
```

3. Write an R program to create and display a 3D array with specified rows, columns, and tables.

```
R Console
> rows <- 2
> columns <- 3
> tables <- 2
>
> # Create the 3D array
> arr <- array(1:(rows * columns * tables),
+             dim = c(rows, columns, tables))
>
> # Display the 3D array
> print("3D Array:")
[1] "3D Array:"
> print(arr)
, , 1
      [,1] [,2] [,3]
[1,]  1    3    5
[2,]  2    4    6
, , 2
      [,1] [,2] [,3]
[1,]  7    9   11
[2,]  8   10   12
>

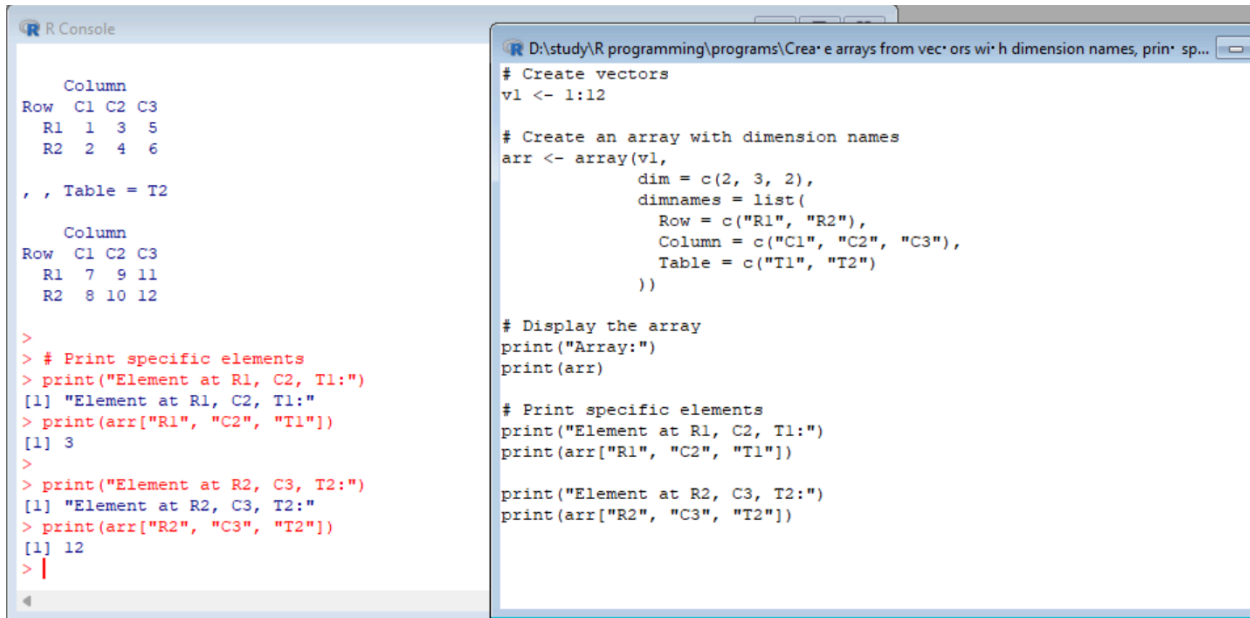
D:\study\R programming\programs\Write an R program to create and display a 3D array with spe...
# Create a 3D array with specified rows, columns, and tables

# Specify dimensions
rows <- 2
columns <- 3
tables <- 2

# Create the 3D array
arr <- array(1:(rows * columns * tables),
             dim = c(rows, columns, tables))

# Display the 3D array
print("3D Array:")
print(arr)
```

4. Create arrays from vectors with dimension names, print specific elements.



The screenshot shows two R console windows. The left window displays the output of a table T2 and the execution of R code to print specific elements from an array. The right window shows the R code to create a 3D array with dimension names and print its elements.

```
R Console
```

```
      Column
Row  C1 C2 C3
R1   1  3  5
R2   2  4  6

, , Table = T2

      Column
Row  C1 C2 C3
R1   7  9 11
R2   8 10 12

>
> # Print specific elements
> print("Element at R1, C2, T1:")
[1] "Element at R1, C2, T1:"
> print(arr["R1", "C2", "T1"])
[1] 3
>
> print("Element at R2, C3, T2:")
[1] "Element at R2, C3, T2:"
> print(arr["R2", "C3", "T2"])
[1] 12
> |
```

```
D:\study\R programming\programs\Create arrays from vectors with dimension names, print sp...
# Create vectors
v1 <- 1:12

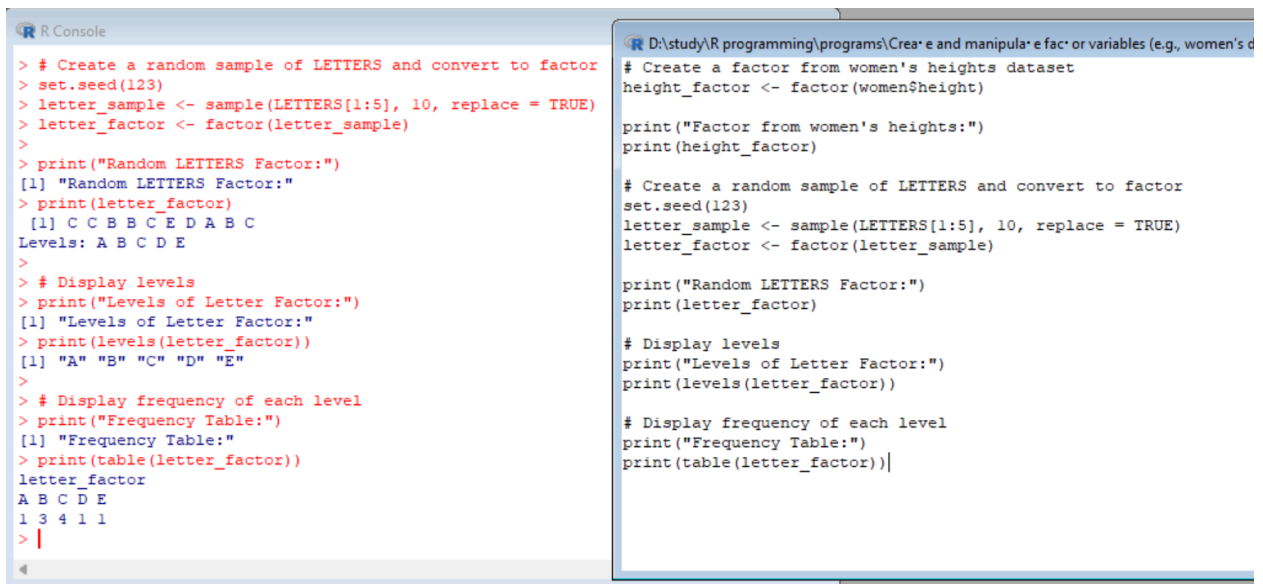
# Create an array with dimension names
arr <- array(v1,
             dim = c(2, 3, 2),
             dimnames = list(
               Row = c("R1", "R2"),
               Column = c("C1", "C2", "C3"),
               Table = c("T1", "T2")
             ))

# Display the array
print("Array:")
print(arr)

# Print specific elements
print("Element at R1, C2, T1:")
print(arr["R1", "C2", "T1"])

print("Element at R2, C3, T2:")
print(arr["R2", "C3", "T2"])
```

5. Create and manipulate factor variables (e.g., women's daase heighs, random LETTERS sample).



The screenshot shows two R console windows. The left window displays the execution of R code to create a random sample of letters and convert it to a factor, then print its levels and frequency table. The right window shows the R code to create a factor from a dataset of women's heights and print its levels and frequency table.

```
R Console
```

```
> # Create a random sample of LETTERS and convert to factor
> set.seed(123)
> letter_sample <- sample(LETTERS[1:5], 10, replace = TRUE)
> letter_factor <- factor(letter_sample)
>
> print("Random LETTERS Factor:")
[1] "Random LETTERS Factor:"
> print(letter_factor)
[1] C C B B C E D A B C
Levels: A B C D E

> # Display levels
> print("Levels of Letter Factor:")
[1] "Levels of Letter Factor:"
> print(levels(letter_factor))
[1] "A" "B" "C" "D" "E"
>
> # Display frequency of each level
> print("Frequency Table:")
[1] "Frequency Table:"
> print(table(letter_factor))
letter_factor
A B C D E
1 3 4 1 1
> |
```

```
D:\study\R programming\programs\Create and manipulate factor variables (e.g., women's d
# Create a factor from women's heights dataset
height_factor <- factor(women$height)

print("Factor from women's heights:")
print(height_factor)

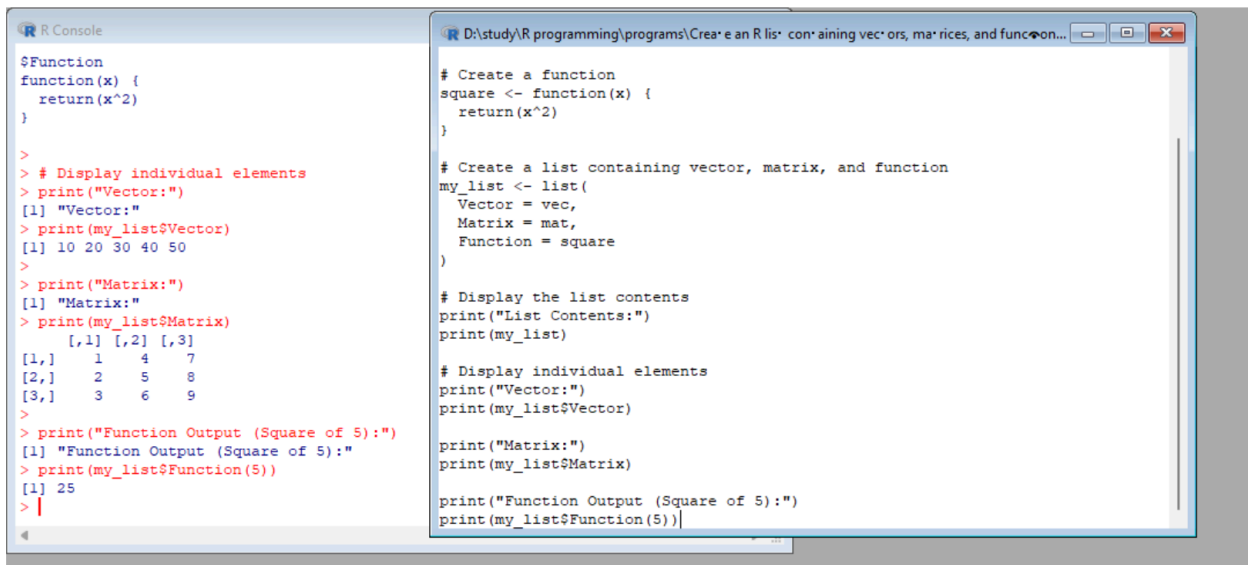
# Create a random sample of LETTERS and convert to factor
set.seed(123)
letter_sample <- sample(LETTERS[1:5], 10, replace = TRUE)
letter_factor <- factor(letter_sample)

print("Random LETTERS Factor:")
print(letter_factor)

# Display levels
print("Levels of Letter Factor:")
print(levels(letter_factor))

# Display frequency of each level
print("Frequency Table:")
print(table(letter_factor))
```

6. Create an R list containing vectors, matrices, and functions; display contents.



The image shows two windows from an R environment. The left window is the R Console, and the right window is an R script editor titled "D:\study\R programming\programs\Create an R list containing vectors, matrices, and functions...".

R Console Output:

```
$Function
function(x) {
  return(x^2)
}

>
> # Display individual elements
> print("Vector:")
[1] "Vector:"
> print(my_list$Vector)
[1] 10 20 30 40 50
>
> print("Matrix:")
[1] "Matrix:"
> print(my_list$Matrix)
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
>
> print("Function Output (Square of 5):")
[1] "Function Output (Square of 5):"
> print(my_list$Function(5))
[1] 25
>
```

R Script Editor Code:

```
# Create a function
square <- function(x) {
  return(x^2)
}

# Create a list containing vector, matrix, and function
my_list <- list(
  Vector = vec,
  Matrix = mat,
  Function = square
)

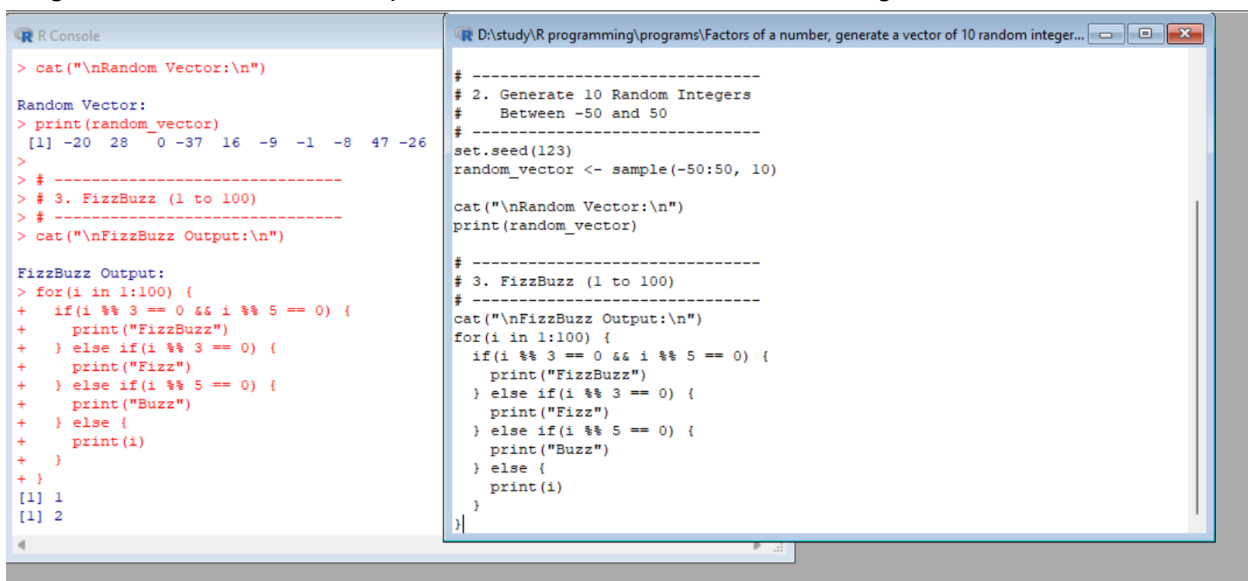
# Display the list contents
print("List Contents:")
print(my_list)

# Display individual elements
print("Vector:")
print(my_list$Vector)

print("Matrix:")
print(my_list$Matrix)

print("Function Output (Square of 5):")
print(my_list$Function(5))
```

7. Write R programs for basic tasks: Factors of a number, generate a vector of 10 random integers between -50 and 50, print numbers 1–100 with FizzBuzz logic.



The image shows two windows from an R environment. The left window is the R Console, and the right window is an R script editor titled "D:\study\R programming\programs\Factors of a number, generate a vector of 10 random integer...".

R Console Output:

```
> cat("\nRandom Vector:\n")

Random Vector:
> print(random_vector)
[1] -20 28 0 -37 16 -9 -1 -8 47 -26
>
> # -----
> # 3. FizzBuzz (1 to 100)
> # -----
> cat("\nFizzBuzz Output:\n")

FizzBuzz Output:
> for(i in 1:100) {
+   if(i %% 3 == 0 && i %% 5 == 0) {
+     print("FizzBuzz")
+   } else if(i %% 3 == 0) {
+     print("Fizz")
+   } else if(i %% 5 == 0) {
+     print("Buzz")
+   } else {
+     print(i)
+   }
+ }
[1] 1
[1] 2
```

R Script Editor Code:

```
# -----
# 2. Generate 10 Random Integers
# Between -50 and 50
# -----
set.seed(123)
random_vector <- sample(-50:50, 10)

cat("\nRandom Vector:\n")
print(random_vector)

# -----
# 3. FizzBuzz (1 to 100)
# -----
cat("\nFizzBuzz Output:\n")
for(i in 1:100) {
  if(i %% 3 == 0 && i %% 5 == 0) {
    print("FizzBuzz")
  } else if(i %% 3 == 0) {
    print("Fizz")
  } else if(i %% 5 == 0) {
    print("Buzz")
  } else {
    print(i)
  }
}
```